The Kalman Filter

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1) The Basics

This project details the use of a Kalman filter for estimating the state of a rocket during flight using noisy sensor data and a simplified physics-based model. By simulating the first one-dimensional stage, the system tracks altitude and velocity based off of an acceleration control input and altitude as a measurement. Realistic sensor noise is used to emulate uncertainties commonly encountered in such scenarios, such as imperfect altimeters and accelerometers. The Kalman filter is then implemented to repeatedly combine model predictions with measurements, producing an optimal estimate of the rocket's state over time. This approach demonstrates how state estimation techniques can significantly improve accuracy compared to simply relying on raw sensor data. This highlights the importance of data fusion in GNC applications.

2) Key Matrices

2.1 – State Vector (x)

The state vector is the estimated state of the rocket given by the Kalman filter. It takes the form of:

$$[1] \qquad x = \begin{bmatrix} height \\ velocity \end{bmatrix} = \begin{bmatrix} h \\ v \end{bmatrix}$$

Initially this will be set to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and future iterations of x will be calculated as:

$$[2] \qquad x_{i+1} = Fx_i + Ba_i$$

Where F is the state transition matrix, B is the control input matrix, and a is the measured acceleration at time i.

2.2 – State Transition Matrix (F)

As we just covered, in equation 2 the state transition matrix is multiplied by x_i, in doing this we are using kinematics to estimate the state at x_{i+1}. F is constructed as shown below:

$$[3] \qquad F = \begin{bmatrix} 1 & dt \\ 0 & 1 \end{bmatrix}$$

Therefore, when F is multiplied by x we are given the following equations:

$$[4] \qquad h_{i+1} = h_i + v_i \cdot dt$$

$$[5] \qquad v_{i+1} = v_i$$

Based off of these equations you can see this is a simplified model, the state transition matrix does not use acceleration to estimate the state. That is left to the control input matrix.

2.3 – Control Input Matrix (B)

In the state vector calculation (equation 2), we multiplied B by the current acceleration (a) we do this in order to account for the acceleration in the state estimation. B is built as shown:

$$[6] \qquad B = \begin{bmatrix} \frac{1}{2}dt^2 \\ dt \end{bmatrix}$$

When multiplied by a we get:

$$[7] \qquad h_{i+1} = \frac{1}{2}a \cdot dt^2$$

$$[8] \qquad v_{i+1} = a \cdot dt$$

Combining the product of the control input matrix (B) and the acceleration vector (a) with the product of the state transition matrix (F) and the current state vector (x_i) is how we get the predictive next step of the state vector (x_{i+1}).

2.4 – Covariance Matrix (P)

The covariance matrix is created by estimating the standard deviation of the error in order to tell the Kalman filter just how much error it should be expecting in the data set. The covariance matrix is a diagonal matrix consisting of the standard deviations squared, (1,1) being the standard deviation of the height error and (2,2) being that of the velocity.

$$[9] \qquad P = \begin{bmatrix} \sigma_h^2 & 0 \\ 0 & \sigma_v^2 \end{bmatrix}$$

This is set initially guessing the standard deviation of the error and is adjusted during every iteration in order to more accurately estimate σ. During these iterations we also introduce the off diagonal as the covariance which represents how error in height and velocity is correlated. We create the future iterations of P with the following equations:

$$[10] \qquad P_{pred} = FPF^T + Q$$

$$[11] \qquad P = (I - KH)P_{pred}$$

Using these equations the covariance matrix takes the form of:

$$[12] \qquad P = \begin{bmatrix} \sigma_h^2 & cov(h,v) \\ cov(v,h) & \sigma_v^2 \end{bmatrix}$$

Using this model the covariance matrix can be interpreted as a measure of the filter's confidence, increasing when relying on predictions and decreasing when incorporating measurements.

2.5 – Process Noise (Q)

Process noise represents the uncertainty of your model this is not an estimate of the error in your data. For example, on this project I modeled the kinematics of the rocket in order to get an idea of the altitude and velocity versus time, however my model is very simple and doesn't consider drag or thrust variation. In reality these are happening and will affect the rocket's altitude and velocity. This is where I need to account for those discontinuities. Because of this we use the process noise in equation 10 when calculating P_{pred} this accounts for potential error in our calculation and creates a more reliable estimate. Q can be estimated as:

$$[13] \qquad Q \propto \begin{bmatrix} \frac{dt^4}{4} & \frac{dt^3}{2} \\ \frac{dt^3}{2} & dt^2 \end{bmatrix}$$

3) Key Values

3.1 – Measurement Noise (R)

Measurement noise is a value showing how much we trust the sensors we are using. This depends on the robustness of the sensors you are using; however, it can generally be estimated that R is similar to the following.

3.2 – Residual Covariance (S)

Residual covariance is based on a different value called the residual; the residual represents the error between the model's estimate and the sensor's reading. The equation is shown below where z is the altitude sensor reading.

$$[14] \qquad y = z - Hx_{pred}$$

The residual covariance comes in to represent how much we trust the residual, the error that the residual represents is distorted due to noise in the sensor as well as model deficiencies that we mentioned earlier such as drag or thrust variation. S is computed as follows:

$$[15] \qquad S = HP_{pred}H^T + R$$

This expands into:

$$[16] \qquad S = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \sigma_h^2 & cov(h,v) \\ cov(v,h) & \sigma_v^2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + R = \sigma_h^2 + R$$

In its simplest form residual covariance represents the sum of model uncertainty and measurement uncertainty.

3.3 – Kalman Gain (K)

The Kalman Gain is the center of the Kalman Filter, it decides how heavily you weigh the sensor measurement as opposed to your model prediction. If you have an outlier sensor read the Kalman gain will adjust and more heavily favor the model's prediction and vice versa. The equation does a good job of explaining how the gain is calculated.

$$K = P_{pred} H^T S^{-1} \tag{17}$$

Here we are taking the uncertainty of the model's prediction and dividing it by the uncertainty of the residual. This is taking model error and dividing by total error, if K is high that means that model error accounts for most of the error in the system and the filter will side with the sensor's reading and vice versa.

4) The Process